

What is encapsulation in java

Encapsulation in Java is a fundamental object-oriented programming (OOP) concept that involves bundling data (variables) and the methods that operate on that data into a single unit, typically a class. It acts as a protective barrier, preventing direct access to an object's internal state and ensuring that data is modified only through well-defined interfaces. This is achieved by declaring class variables as `private` and providing `public` getter and setter methods to access and modify them. Encapsulation enhances security, provides better control over data, promotes code maintainability, and offers flexibility in code evolution. It is distinct from abstraction, though they complement each other in software design.